

BG77xA-GL&BG95xA-GL

PPP Application Note

LPWA Module Series

Version: 1.0

Date: 2023-02-13

Status: Released



At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

Quectel Wireless Solutions Co., Ltd.

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: info@quectel.com

Or our local offices. For more information, please visit:

<http://www.quectel.com/support/sales.htm>.

For technical support, or to report documentation errors, please visit:

<http://www.quectel.com/support/technical.htm>.

Or email us at: support@quectel.com.

Legal Notices

We offer information as a service to you. The provided information is based on your requirements and we make every effort to ensure its quality. You agree that you are responsible for using independent analysis and evaluation in designing intended products, and we provide reference designs for illustrative purposes only. Before using any hardware, software or service guided by this document, please read this notice carefully. Even though we employ commercially reasonable efforts to provide the best possible experience, you hereby acknowledge and agree that this document and related services hereunder are provided to you on an “as available” basis. We may revise or restate this document from time to time at our sole discretion without any prior notice to you.

Use and Disclosure Restrictions

License Agreements

Documents and information provided by us shall be kept confidential, unless specific permission is granted. They shall not be accessed or used for any purpose except as expressly provided herein.

Copyright

Our and third-party products hereunder may contain copyrighted material. Such copyrighted material shall not be copied, reproduced, distributed, merged, published, translated, or modified without prior written consent. We and the third party have exclusive rights over copyrighted material. No license shall be granted or conveyed under any patents, copyrights, trademarks, or service mark rights. To avoid ambiguities, purchasing in any form cannot be deemed as granting a license other than the normal non-exclusive, royalty-free license to use the material. We reserve the right to take legal action for noncompliance with abovementioned requirements, unauthorized use, or other illegal or malicious use of the material.

Trademarks

Except as otherwise set forth herein, nothing in this document shall be construed as conferring any rights to use any trademark, trade name or name, abbreviation, or counterfeit product thereof owned by Quectel or any third party in advertising, publicity, or other aspects.

Third-Party Rights

This document may refer to hardware, software and/or documentation owned by one or more third parties ("third-party materials"). Use of such third-party materials shall be governed by all restrictions and obligations applicable thereto.

We make no warranty or representation, either express or implied, regarding the third-party materials, including but not limited to any implied or statutory, warranties of merchantability or fitness for a particular purpose, quiet enjoyment, system integration, information accuracy, and non-infringement of any third-party intellectual property rights with regard to the licensed technology or use thereof. Nothing herein constitutes a representation or warranty by us to either develop, enhance, modify, distribute, market, sell, offer for sale, or otherwise maintain production of any our products or any other hardware, software, device, tool, information, or product. We moreover disclaim any and all warranties arising from the course of dealing or usage of trade.

Privacy Policy

To implement module functionality, certain device data are uploaded to Quectel's or third-party's servers, including carriers, chipset suppliers or customer-designated servers. Quectel, strictly abiding by the relevant laws and regulations, shall retain, use, disclose or otherwise process relevant data for the purpose of performing the service only or as permitted by applicable laws. Before data interaction with third parties, please be informed of their privacy and data security policy.

Disclaimer

- a) We acknowledge no liability for any injury or damage arising from the reliance upon the information.
- b) We shall bear no liability resulting from any inaccuracies or omissions, or from the use of the information contained herein.
- c) While we have made every effort to ensure that the functions and features under development are free from errors, it is possible that they could contain errors, inaccuracies, and omissions. Unless otherwise provided by valid agreement, we make no warranties of any kind, either implied or express, and exclude all liability for any loss or damage suffered in connection with the use of features and functions under development, to the maximum extent permitted by law, regardless of whether such loss or damage may have been foreseeable.
- d) We are not responsible for the accessibility, safety, accuracy, availability, legality, or completeness of information, advertising, commercial offers, products, services, and materials on third-party websites and third-party resources.

Copyright © Quectel Wireless Solutions Co., Ltd. 2023. All rights reserved.

About the Document

Revision History

Version	Date	Author	Description
-	2023-02-13	Abel ZHOU	Creation of the document
1.0	2023-02-13	Abel ZHOU	First official release

Contents

About the Document.....	3
Contents	4
Table Index.....	6
Figure Index	7
1 Introduction	8
1.1. Applicable Modules	8
2 Preparation and Settings for PPP	9
2.1. Overview	9
2.2. Port Setting	12
2.3. RK Installation	13
3 PPP Low-speed Dial-up.....	14
3.1. Preparations for Low-speed Dial-up	14
3.2. Modem Configuration.....	15
3.2.1. Add a New Modem.....	15
3.2.2. Configure the Modem Driver	18
3.3. Dial-up Network Configuration	18
3.3.1. Set up a New Connection	18
3.3.2. Configure the Connection	19
3.3.3. Configure the Dial-up Tool.....	21
3.4. Establish Dial-up Connection.....	23
4 PPP High-speed Dial-up.....	24
4.1. Preparations for High-speed Dial-up	24
4.2. Modem Configuration.....	25
4.2.1. Add a New Modem.....	25
4.2.2. Configure the Modem Driver	26
4.3. Dial-up Network Configuration	27
4.3.1. Set up a New Connection	27
4.3.2. Configure the Connection	27
4.3.3. Configure the Dial-up Tool.....	27
4.4. Establish the Dial-up Connection.....	28
4.5. Restore Default Port Mapping Functions.....	28
5 Route Configuration	29
5.1. Query IP Address	29
5.2. Delete All Default Routes	29
5.3. Add Target Routes	29
6 FAQ.....	30
6.1. Disable Driver Signature Enforcement	30
6.1.1. Disable the Driver Signature Enforcement	31
6.1.2. Update Driver	33

7	Appendix References	35
---	---------------------------	----

Table Index

Table 1: Applicable Modules.....	8
Table 2: Key Switchers and Ports of BG77xA-GL TE-A and BG95xA-GL TE-A.....	10
Table 3: Port Mapping Function of TE-A (Debug Mode is Enabled with AT+QCFGEXT="debug",1).....	12
Table 4: Port Mapping Function of TE-A.....	24
Table 5: Related Documents	35
Table 6: Terms and Abbreviations	35

Figure Index

Figure 1: BG770A-GL TE-A	9
Figure 2: BG950A-GL TE-A	10
Figure 3: Ports in Device Management.....	12
Figure 4: Check RK Version.....	13
Figure 5: Search Phone and Modem.....	15
Figure 6: Add a New Modem.....	16
Figure 7: Install a New Modem	17
Figure 8: Select a Port to Install the Modem – Low-Speed Dial-up.....	17
Figure 9: Set Modem Properties - Low-Speed Dial-up.....	18
Figure 10: Set up a New Connection or Network	19
Figure 11: Set up a Dial-up Connection.....	19
Figure 12: Type the Information for Setting up a Dial-up Connection	20
Figure 13: Reconfigure the Connection	21
Figure 14: Change Adapter Settings.....	21
Figure 15: Select Connection Properties	22
Figure 16: General Settings for Modem Configuration - Low-Speed Dial-up.....	22
Figure 17: Establish the Dial-up Connection	23
Figure 18: Install 921600 Modem Driver.....	25
Figure 19: Select a Port to Install the Modem - High-Speed Dial-up.....	26
Figure 20: Set Modems Properties - High-Speed Dial-up	27
Figure 21: General Settings for Modem Configuration - High-Speed Dial-up.....	28
Figure 22: Set Modem Properties	30
Figure 23: Open "Update & Security"	31
Figure 24: Restart Now	31
Figure 25: Change Windows Startup Behaviour.....	32
Figure 26: Disable the Driver Signature Enforcement	32
Figure 27: Find Driver Forder Path	33
Figure 28: Update Drivers	33
Figure 29: Update Drivers Success	34
Figure 30: Set Modem Properties	34

1 Introduction

The Point-to-Point Protocol (PPP) is designed for simple links which transport packets between two ports. These links provide full-duplex simultaneous bi-directional operation, and are assumed to deliver packets in order. It is intended that PPP provides a common solution for easy connection of a wide variety of hosts, bridges and routers.

This document explains how to use PPP function on Quectel BG77xA-GL or BG95xA-GL TE-A, including preparation and settings for PPP and host configuration for PPP dial-up.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
BG77xA-GL	BG770A-GL
	BG772A-GL
	BG773A-GL
BG95xA-GL	BG950A-GL
	BG951A-GL
	BG952A-GL
	BG953A-GL
	BG955A-GL

2 Preparation and Settings for PPP

This document introduces PPP dial-up operations based on BG77xA-GL or BG95xA-GL TE-A. If customer devices are used instead of the TE-A, make sure that the interfaces listed in **Table 1** are also available on customer devices. Otherwise, contact Quectel Technical Support for development considerations.

2.1. Overview

1. Connect the TE-A to PC with 4 cables (the interfaces marked 5, 8, 9 and 10 in the figure below).

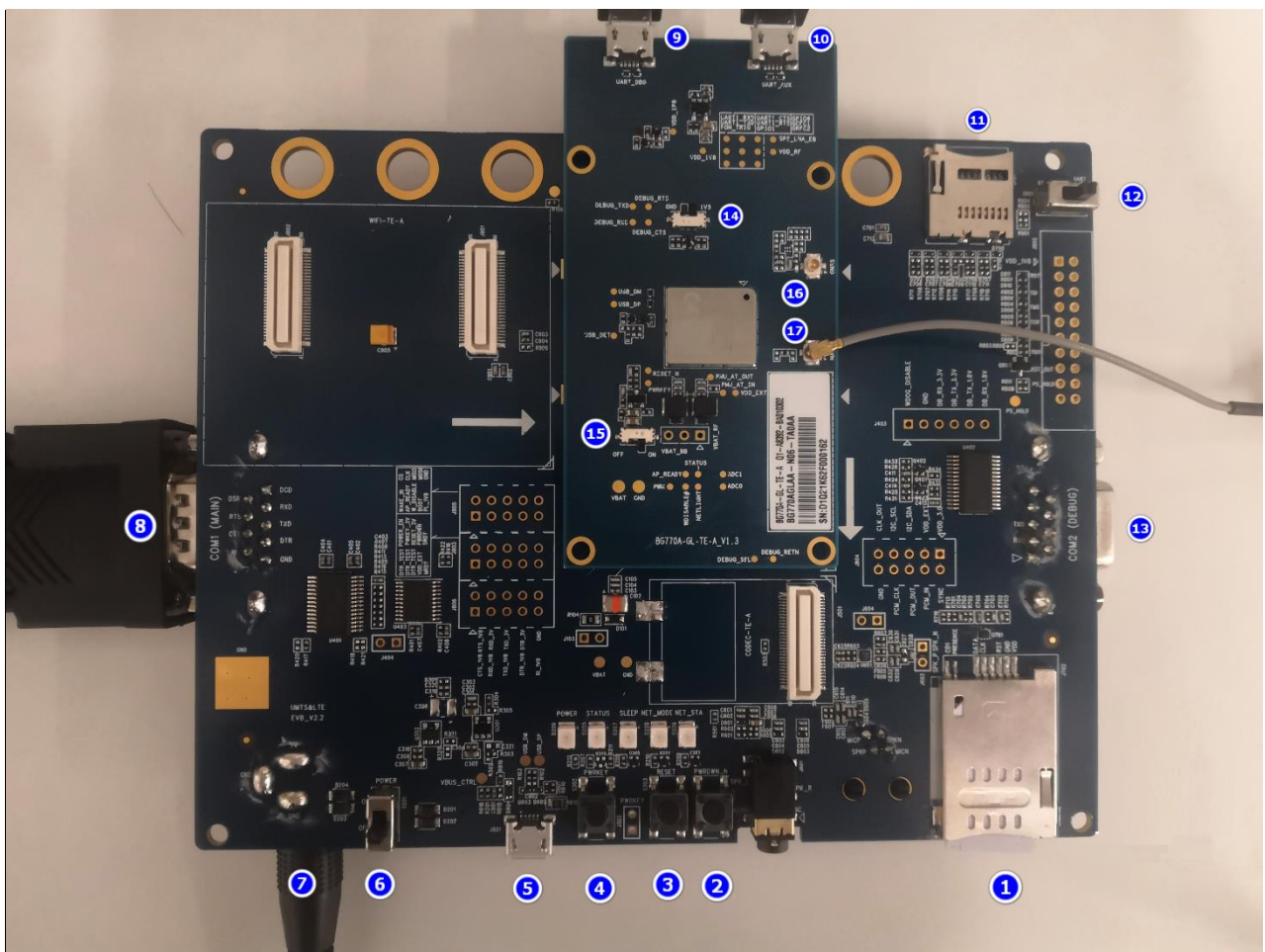


Figure 1: BG770A-GL TE-A

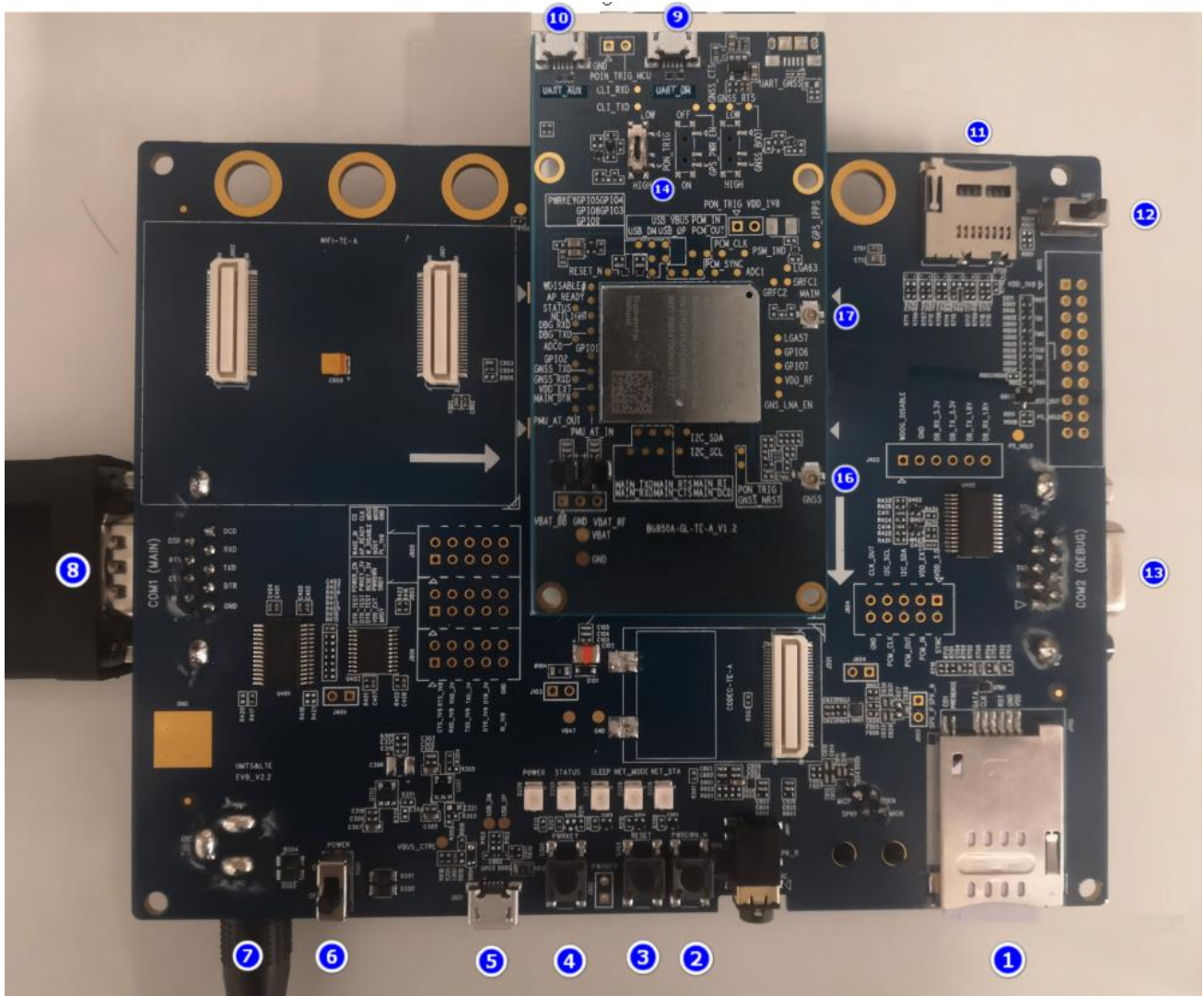


Figure 2: BG950A-GL TE-A

Table 2: Key Switchers and Ports of BG77xA-GL TE-A and BG95xA-GL TE-A

SN	Interface	Description
1	USIM	Only 1.8 V (U)SIM card is supported.
2	PWRDWN_N Button	Reserved.
3	RESET Button	Reset the module.
4	PWRKEY Button	Turn on/off the module.
5	USB	USB 2.0 (FS only)
6	POWER Switcher	Power on/off the module.

7	Power Supply Interface	Supply voltage: 5 V.
8	COM1 (MAIN)	Used for AT command communication. Baud rate: 115200 bps baud by default. Support RTS and CTS hardware flow control.
9	UART_DBG (BG77xA-GL) UART_DM (BG95xA-GL)	Used for firmware upgrade, software debugging, log output and NMEA sentences output. 115200 bps baud rate by default.
10	UART_AUX (BG77xA-GL) UART_AUX (BG95xA-GL)	Used for RF calibration and log output. 921600 bps baud rate by default.
11	MEMORY Card	Used for plugging SD card.
12	UART Switcher	Used for selecting communication object of main UART: "BT" or "UART_PCM". Make sure the switcher is switched to "UART_PCM".
13	QCOM2 (DEBUG)	Reserved.
14	PON_TRIG Switch	If the switch is at "1V8" state, PON_TRIG is at high level; while "GND" state is at low level.
15	Automatic Power-on Switch (BG95xA-GL does not has this interface)	Switching to "ON", the power-on circuit is turned on automatically. When VBAT rushes from 0 V to 3.3 V, the module boots automatically. Switching to "OFF", the power circuit is disconnected automatically. When the automatic power-on circuit is enabled, there is a constant current consumption of about 700 μ A.
16	GNSS	GNSS antenna connector.
17	Main	Main antenna connector.

NOTE

1. For more details of TE-A interfaces, see **documents [1]** and **[2]**.
2. Before connecting the TE-A to PC, please install the UART driver first. For more details, please contact Quectel Technical Support.
3. The differences between BG95xA-GL TE-A and BG77xA-GL TE-A are the location and name of the interfaces (i.e., the interfaces marked 9, 10). See **Figures 1** and **2**.

2.2. Port Setting

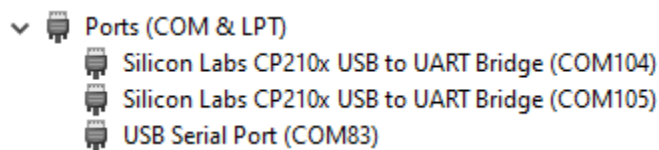


Figure 3: Ports in Device Management

Upon installing the UART driver mentioned in Chapter 2.1, 3 ports like above are shown in **Figure 3**. The port mapping results can be checked by connecting the TE-A to PC and testing the ports one by one.

Table 3: Port Mapping Function of TE-A (Debug Mode is Enabled with AT+QCFGEXT="debug",1)

Port Name	Port ID	Default Usage
COM1 (MAIN)	83	AT_PPP_AUX
UART_AUX	105	FW_LOG
UART_DBG	104	SFP_LOGGER

NOTE

For more details of **AT+QCFGEXT="debug"**, see *document [4]*.

2.3. RK Installation

Step 1: Query the module RK version by executing **AT+QCFGEXT="ver"** (see *document [4]*).

The first line in the response includes the corresponding RK version.

```
AT+QCFGEXT="VER"
NP Package: RK_03_00_00_00_20341_001
Apps: RKAPP_03_00_00_00_20341_001_19_Jan_22_00:04:43
Using APP processor - no SB or 3B versions
MAC Revision: REL_1250_03_00_00_REV_296060
MAC Package Version: ALT1250_03_00_00_00_20331_Fw
MAC Build Time: Jan_18_2022_00_42_28
PHY Revision: 12.50.296039
PHY Build Time: Jan_18_2022_00_09_56
PHY Build Info: releas_0
PMP Revision: 296063
PMP Version: 20331_PMP
PMP build time: Sep_20_2021_18_26_27
DSP Revision: 65262
BB Product: 1250
BB HW Revision: 20
RFIC_6200 Revision: 00
NP Build Time: Mar_2_2022_14:15:35
C&V Ver: 9.62
Calibration date: 25.04.22
Calibrated NP package: RK_03_00_00_00_02
```

Figure 4: Check RK Version

Step 2: Get the RK package named *RK_03_<RK_version>.exe* from Quectel. Then install RK into the default path before logging and using PPP-related features.

3 PPP Low-speed Dial-up

You can perform a low-speed dial-up in the scenario that there is no requirement for speed through all the three ports (see **Figure 3**) at 115200 bps in Cat-M and NB-IoT. This chapter takes BG770A-GL TE-A COM1 (MAIN) as an example to show how to perform a PPP low-speed dial-up.

3.1. Preparations for Low-speed Dial-up

1. Execute **AT+QCFGEXT="debug",1** (see **document [4]**) to enable debug mode.
2. Execute **AT+IPR?** (see **document [3][4]**) to query baud rate of COM1 (MAIN) port mapping AT_PPP_AUX function.
If the returned baud rate is not 115200 bps, execute **AT+IPR=115200;&W** (see **document [3][4]**) to set it to 115200 bps and store the current setting.
3. Execute **AT+QCFGEXT="FTP_SERVER"** (see **document [5][4]**) to query whether FTP server is enabled.
If FTP server is not enabled, execute **AT+QCFGEXT="FTP_SERVER",1** to enable FTP server.
4. Execute **AT+IFC=2,2** (see **document [3][4]**) to enable hardware flow control.
5. Execute **AT+CGATT?** (see **document [3][4]**) to check whether the module is registered on the EPS network successfully. If **+CGATT: 1** is returned, it means that the module is registered on EPS network successfully.
6. Execute **AT+CGPADDR** to query the IP address (see **document 错误!未找到引用源。**).

NOTE

1. Before executing **AT+IFC=2,2**, check "RTS" option in AT tool.
2. The configurations of **AT+IFC** are not saved automatically, so execute **AT+IFC=2,2** again once the module is rest.

3.2. Modem Configuration

3.2.1. Add a New Modem

Step 1: Search "Phone and Modem" in the search box shown below.

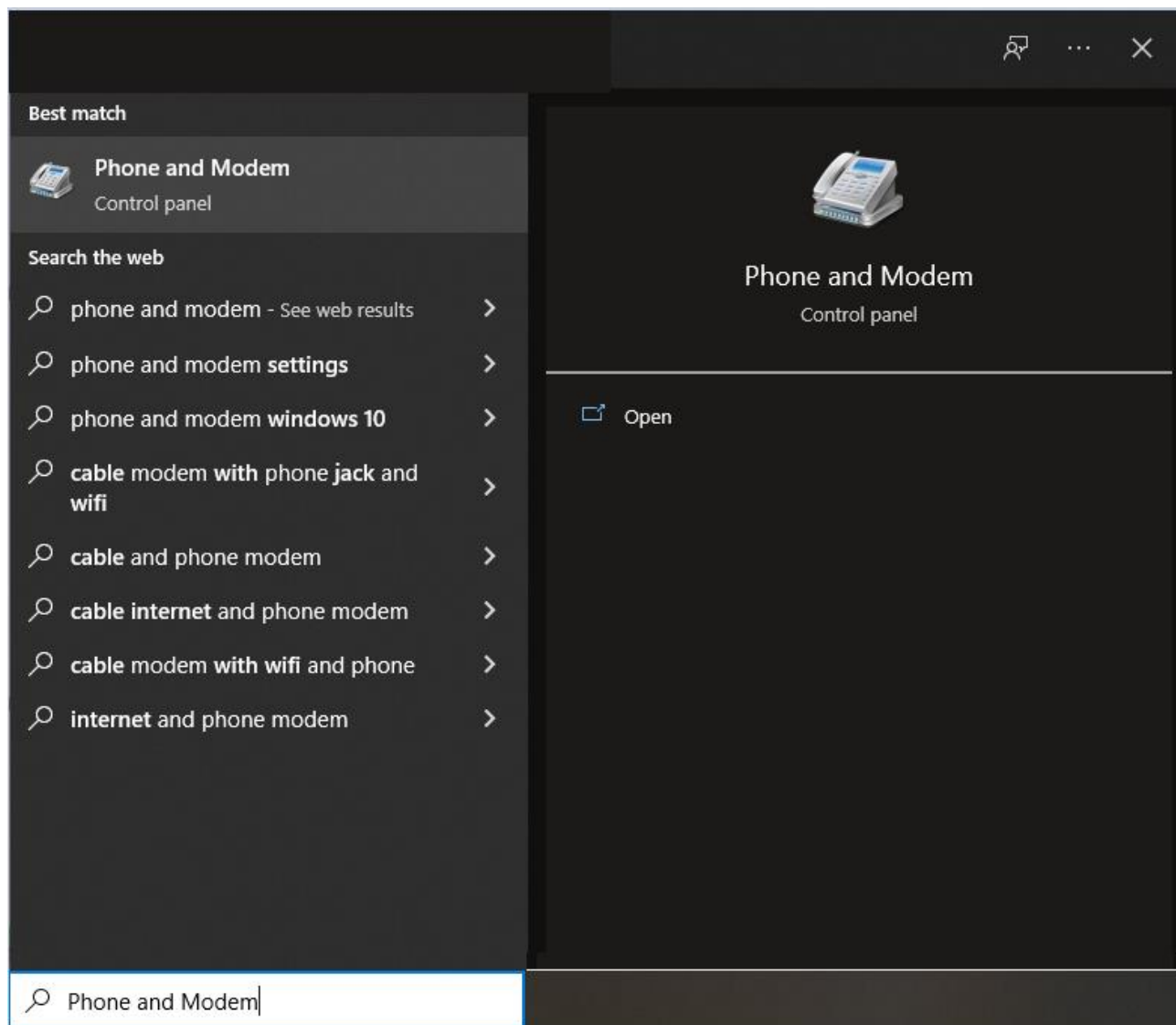


Figure 5: Search Phone and Modem

Step 2: Click "Phone and Modem", click "Modems" -> "Add..." as shown below.

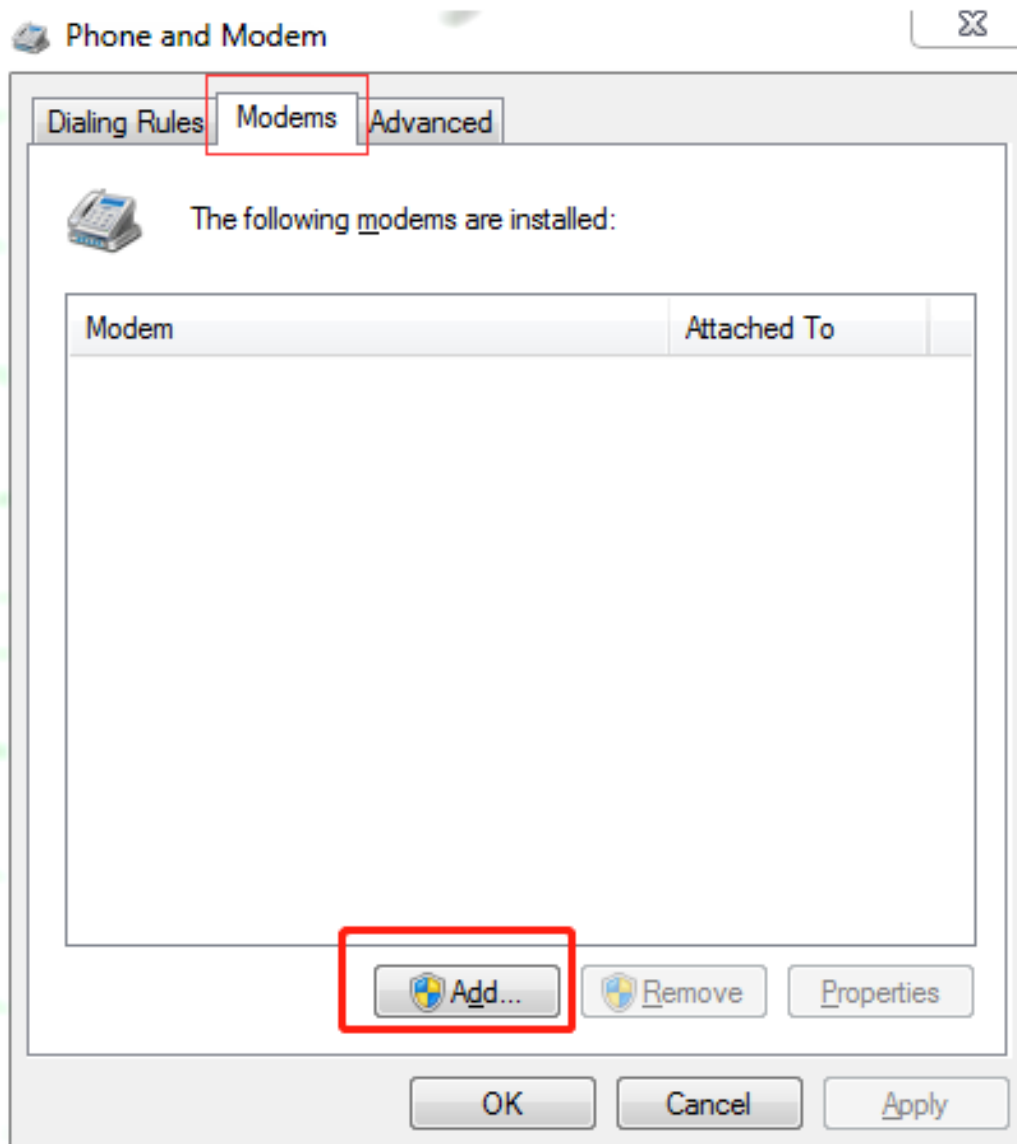


Figure 6: Add a New Modem

Step 3: Check the item "Don't detect my modem; I will select it from a list" and click "Next" to add a new modem.

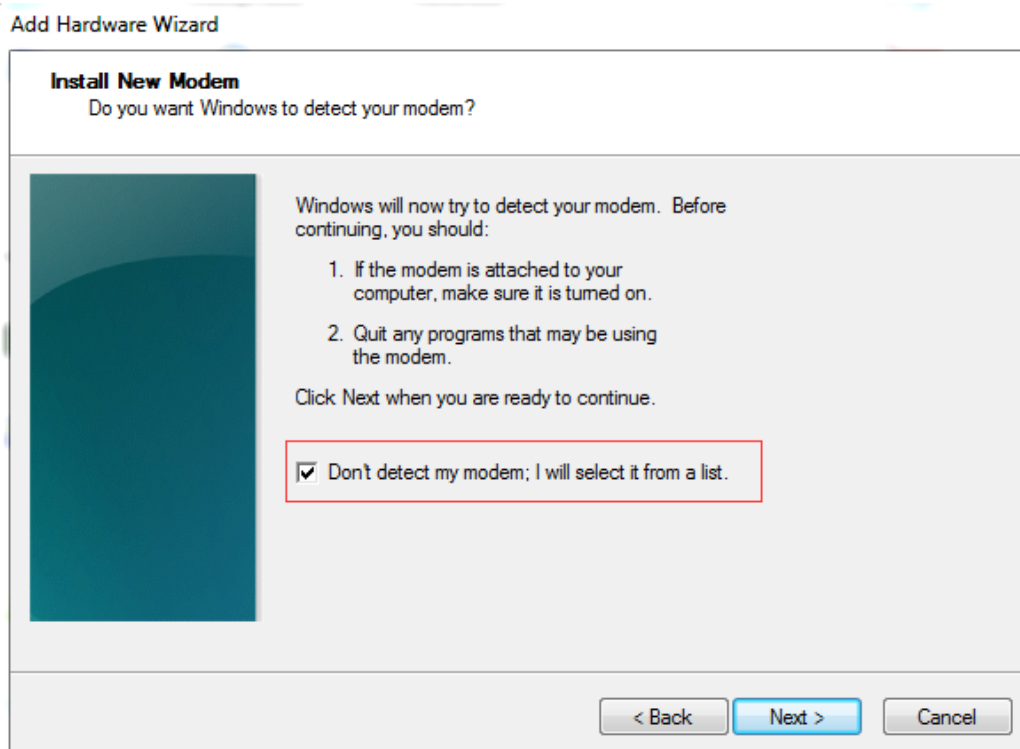


Figure 7: Install a New Modem

Step 4: Refer to the following three figures to install the new modem according to the instructions on the screen: select "Standard 28800 bps Modem", click "Next", select "COM83", click "Next" -> "Finish".

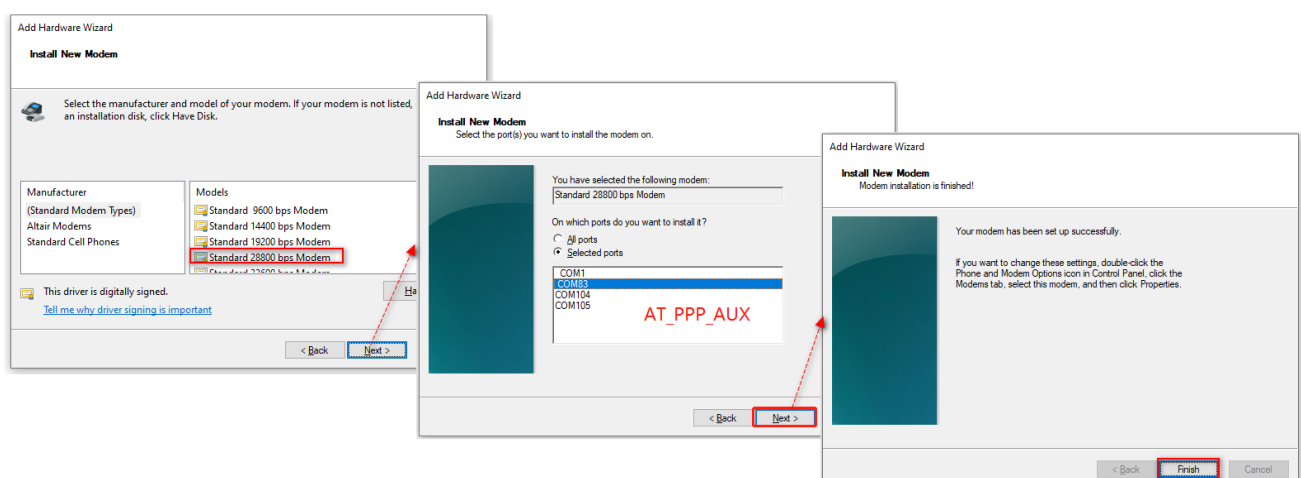


Figure 8: Select a Port to Install the Modem – Low-Speed Dial-up

NOTE

If it fails for COM83 port is unavailable in Figure 8, Make sure that COM83 port (AT port) is not connected to a serial port tool.

3.2.2. Configure the Modem Driver

Step 1: Select the "Standard 28800 bps Modem" which has been installed; click "**Properties**".

Step 2: Click "**Modem**" to select "115200" for "**Maximum Port Speed**"; click "**OK**".

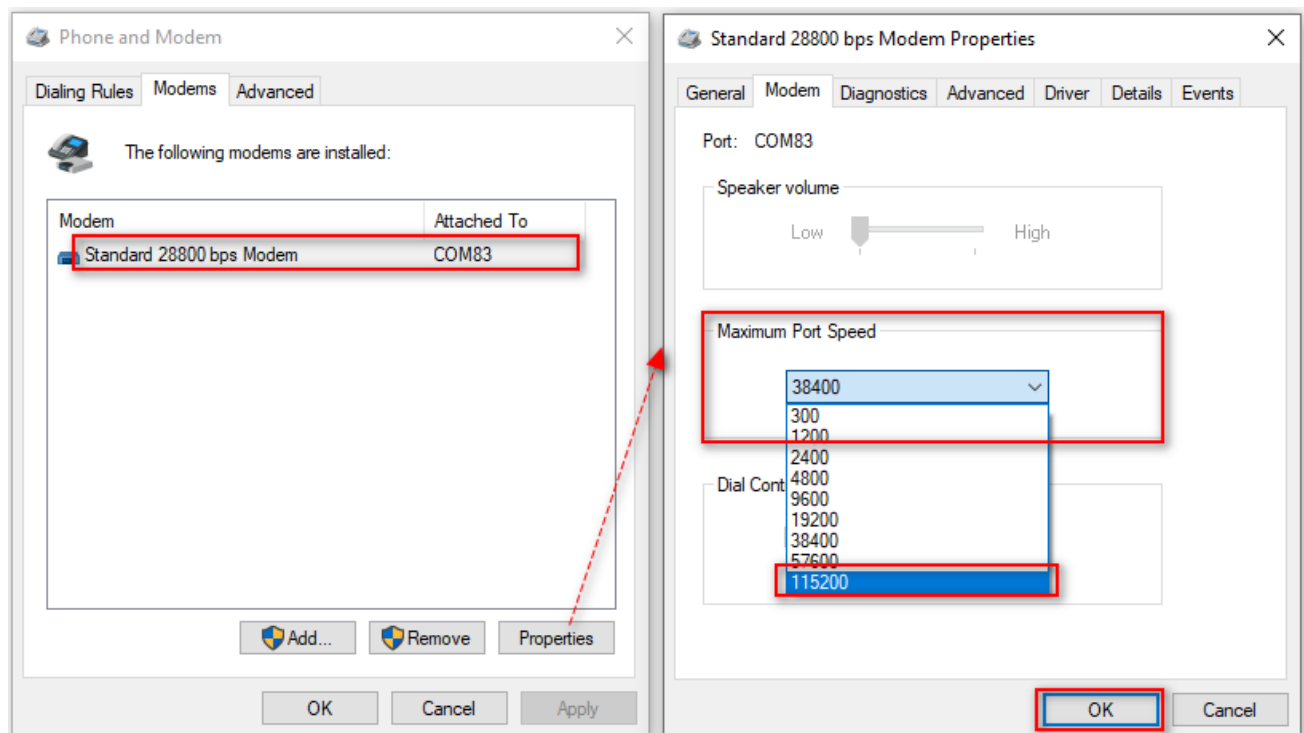


Figure 9: Set Modem Properties - Low-Speed Dial-up

3.3. Dial-up Network Configuration

3.3.1. Set up a New Connection

Open "Control Panel" and click "Network and Internet" -> "Network and Sharing Center" -> "Set up a new connection or network", as illustrated below.

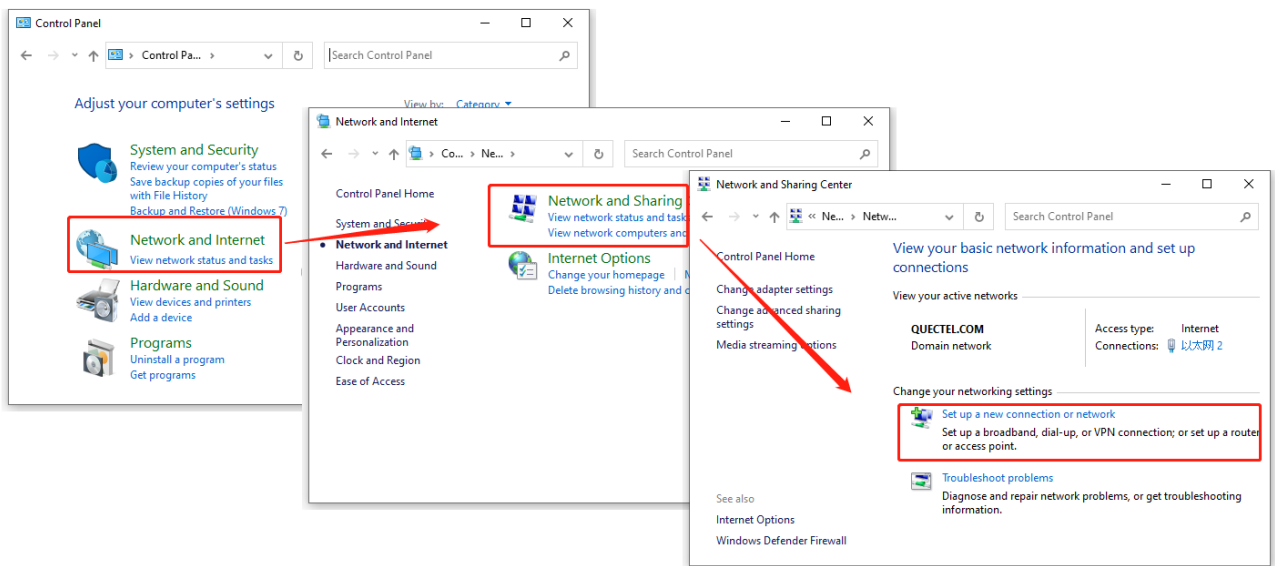


Figure 10: Set up a New Connection or Network

3.3.2. Configure the Connection

Step 1: Select "Connect to Internet" -> "Set Up a New Connection anyway" -> "Dial-up".

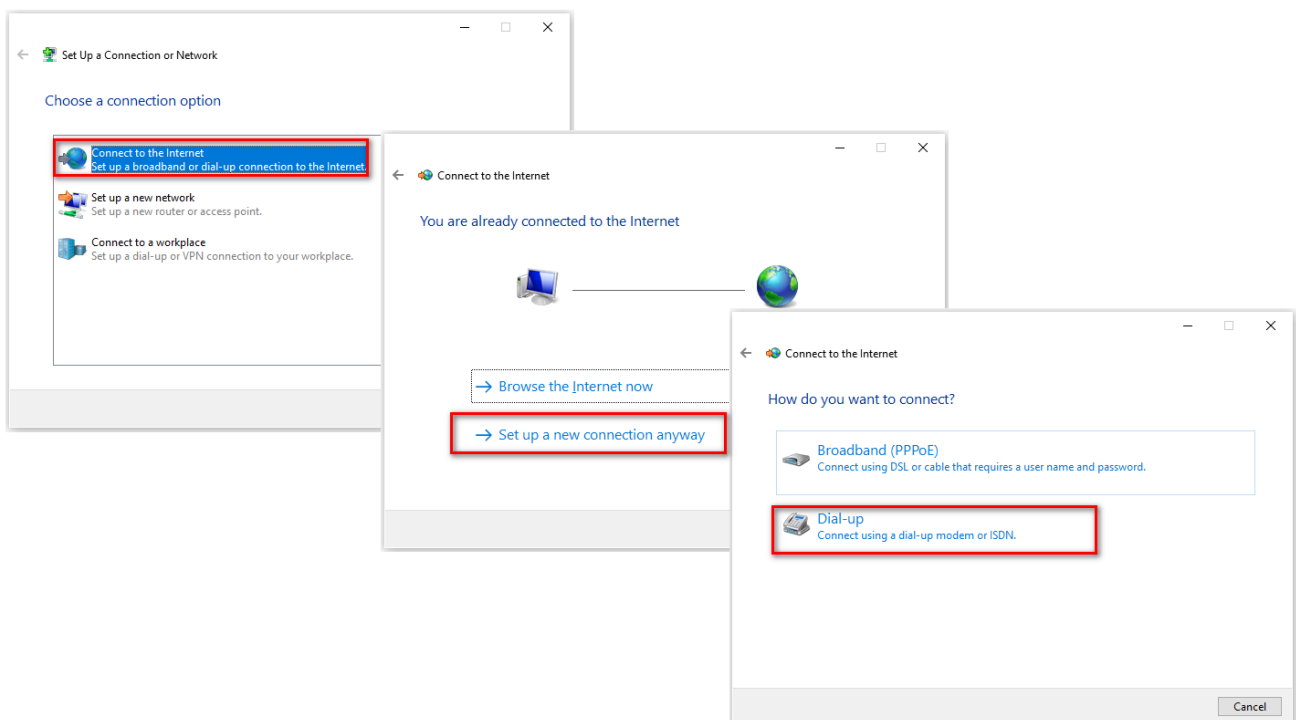
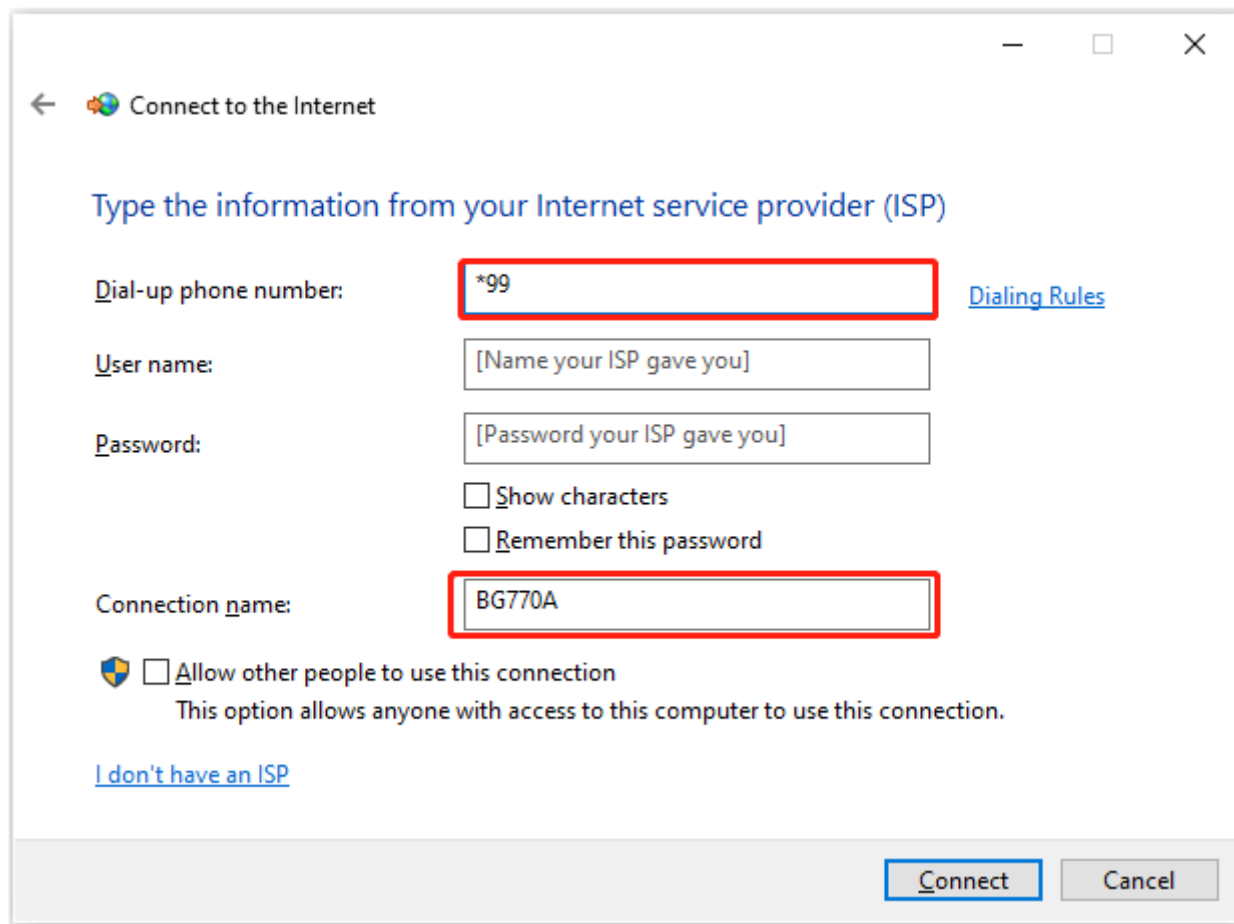


Figure 11: Set up a Dial-up Connection

Step 2: Fill "Dial-up phone number:" with a string "*99", "Connection name" with a proper name "BG770A" as an example in this document, and then click "Connect".



←  Connect to the Internet

Type the information from your Internet service provider (ISP)

Dial-up phone number: [Dialing Rules](#)

User name:

Password:

☐ Show characters

☐ Remember this password

Connection name:

 ☐ Allow other people to use this connection
This option allows anyone with access to this computer to use this connection.

[I don't have an ISP](#)

Figure 12: Type the Information for Setting up a Dial-up Connection

Step 3: If all above settings are correct, skip this step. Otherwise select **"Set up the connection anyway"**, as illustrated below.

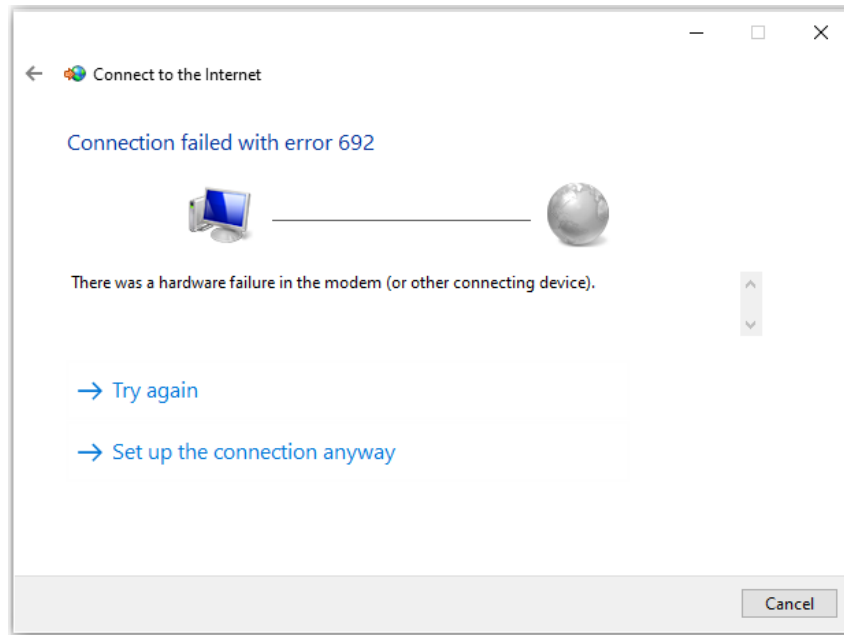


Figure 13: Reconfigure the Connection

3.3.3. Configure the Dial-up Tool

Step 1: Go to **"Change adapter settings"** in **"Control Pane"** -> **"Network and Internet"** -> **"Network and Sharing Center"**.

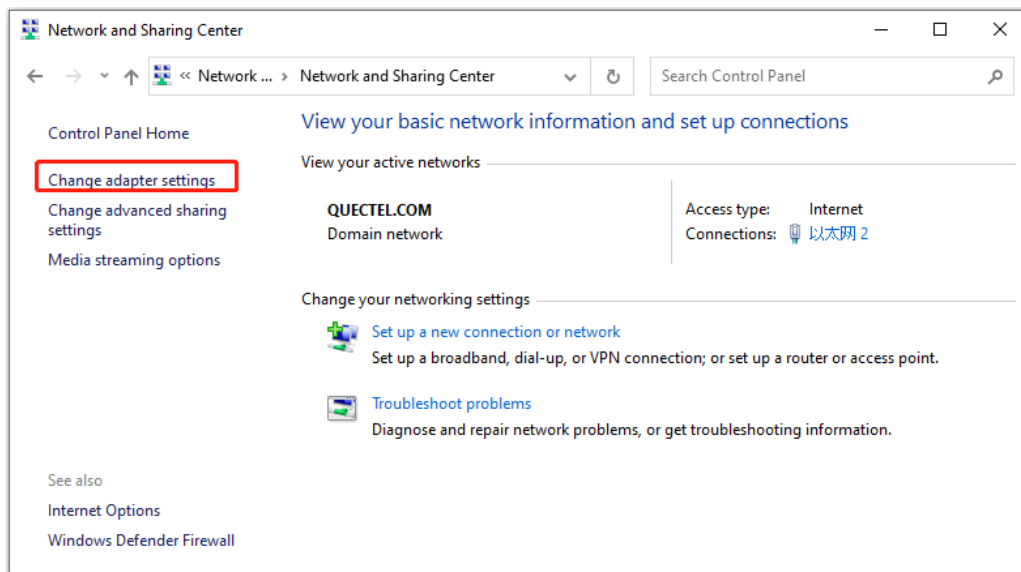


Figure 14: Change Adapter Settings

Step 2: Find the newly created connection and set **"Properties"**.

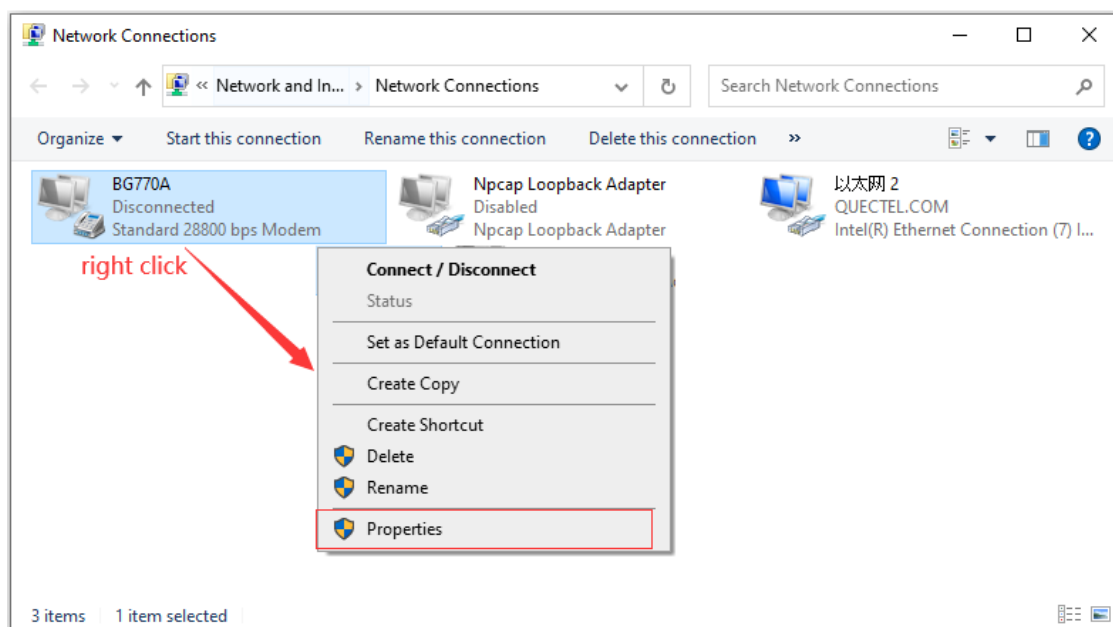


Figure 15: Select Connection Properties

Step 3: Go to **"General"** and click **"Configure..."**, select **"115200"** as **"Maximum speed (bps)"** and then click **"OK"**.

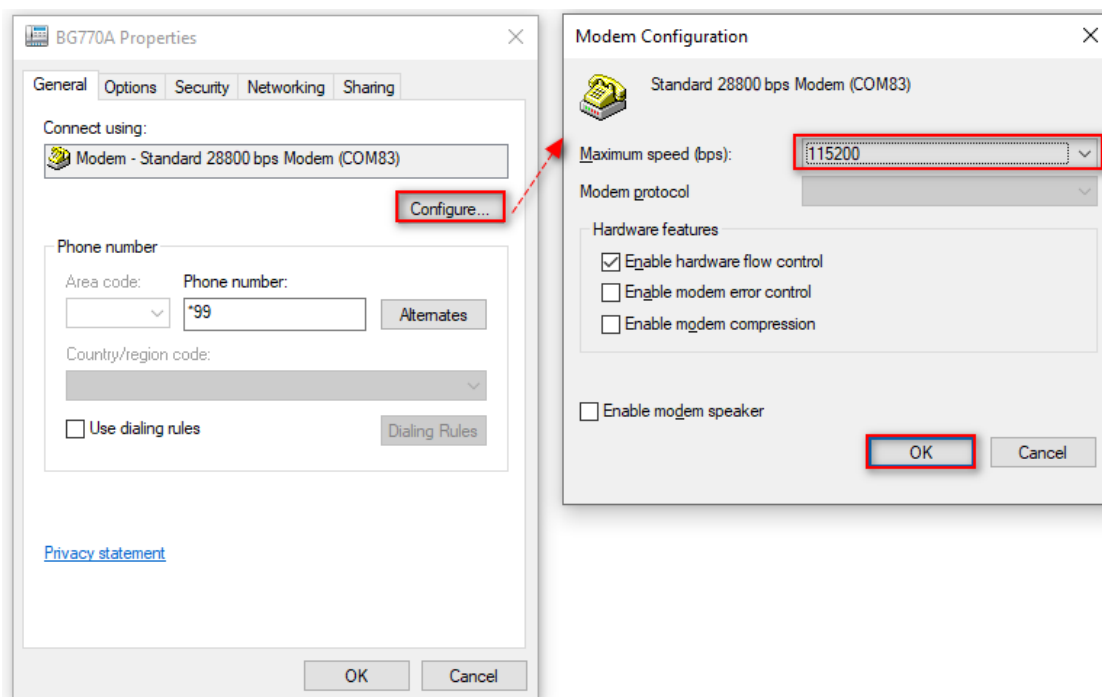


Figure 16: General Settings for Modem Configuration - Low-Speed Dial-up

3.4. Establish Dial-up Connection

Click "**Internet Access**" to find the newly created connection "**BG770A**". Click "**Connect**" and then click "**Dial**" to establish the dial-up connection. Refer to the following figure for details.

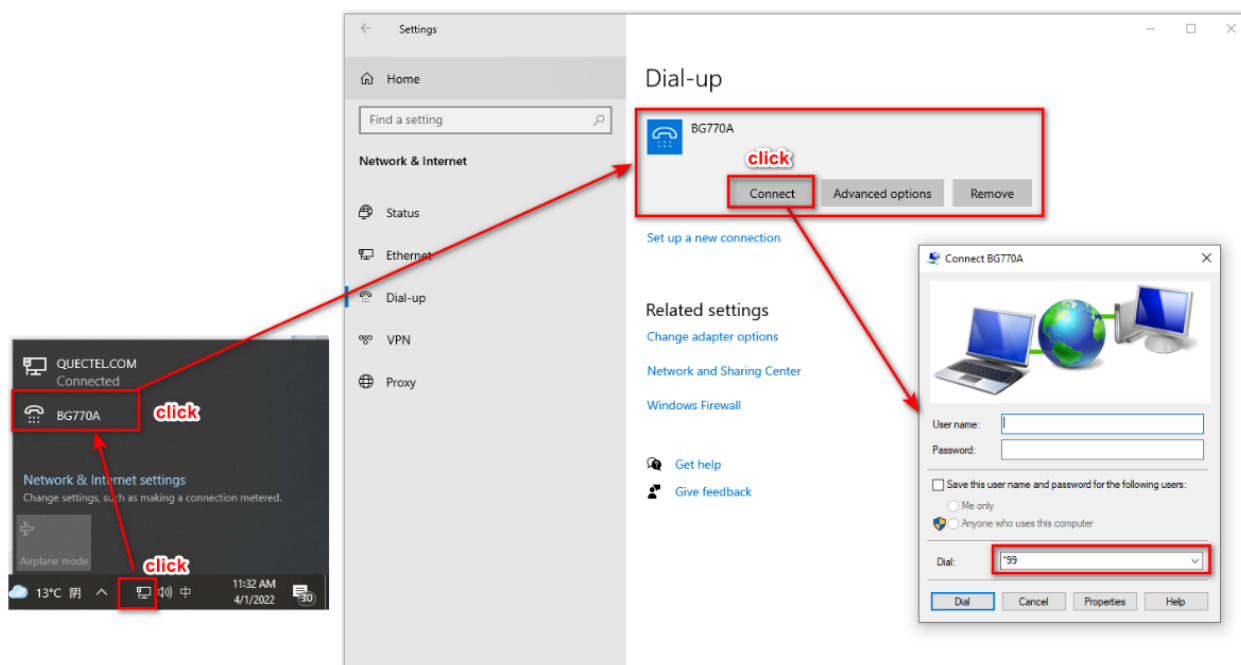


Figure 17: Establish the Dial-up Connection

4 PPP High-speed Dial-up

You can perform a high-speed dial-up in the scenario that there are requirements for speed through COM105, i.e. UART_AUX (see **Figure 3**), at 921600 bps. This chapter takes BG770A-GL TE-A as an example to present you with the steps of performing high-speed dial-up.

4.1. Preparations for High-speed Dial-up

1. Execute **AT+QCFGEXT="debug",0** (see **document [4]**) to disable debug mode.
2. Execute **AT+QCFGEXT="PPPMAPPING/SUBPROFILE",1** (see **document [4]**) to enable PPP mapping.
3. Execute **AT+QCFGEXT="debug",1** to enable debug mode.

At this time, the port mapping function details are as below:

Table 4: Port Mapping Function of TE-A

Port Name	Port ID	Default Usage
COM1 (MAIN)	83	FW_LOG
UART_AUX	105	AT_PPP_AUX
UART_DBG	104	SFP_LOGGER

4. Reboot the module to make the above configuration take effect.
5. Execute **AT+QCFGEXT="FTP_SERVER"** (see **document [5]**) to query whether FTP server is enabled.
If FTP server is not enabled, execute **AT+QCFGEXT="FTP_SERVER",1** to enable FTP server.
6. Execute **AT+IFC=2,2** (see **document [3][4]**) to enable hardware flow control.
7. Execute **AT+IPR?** (see **document [3][4]**) to query baud rate of UART_AUX port mapping AT_PPP_AUX function. If the returned baud rate is not 921600 bps, execute **AT+IPR=921600;&W**

(see **document [3][4]**) to set it to 921600 bps and store the current setting.

8. Execute **AT+CGATT?** (see **document [3][4]**) to check whether the module registered on the EPS network successfully. If **+CGATT: 1** is returned, it means that the module is registered on EPS network successfully.
9. Execute **AT+CGPADDR** to query the IP address (see **document 错误!未找到引用源。**).

NOTE

1. Before executing **AT+IFC=2,2**, check "RTS" option in AT tool.
2. The configuration of **AT+IFC** is not saved automatically, so execute **AT+IFC=2,2** again once the module is reset.
3. Debug mode should be disabled with **AT+QCFGEXT="debug",0** before executing **AT+QCFGEXT="PPPMAPPING/SUBPROFILE",1**.

4.2. Modem Configuration

4.2.1. Add a New Modem

Step 1: Repeat Step 1-3 in **Chapter 3.2.1** to enter "Add Hardware Wizard" window.

Step 2: Refer to the following figures to install the new modem according to the instructions on the screen: select "**Communication cable between two computers**", click "**Have Disk...**" -> "**Browse**", and select "**altair_921600_modem**" to complete the installation.

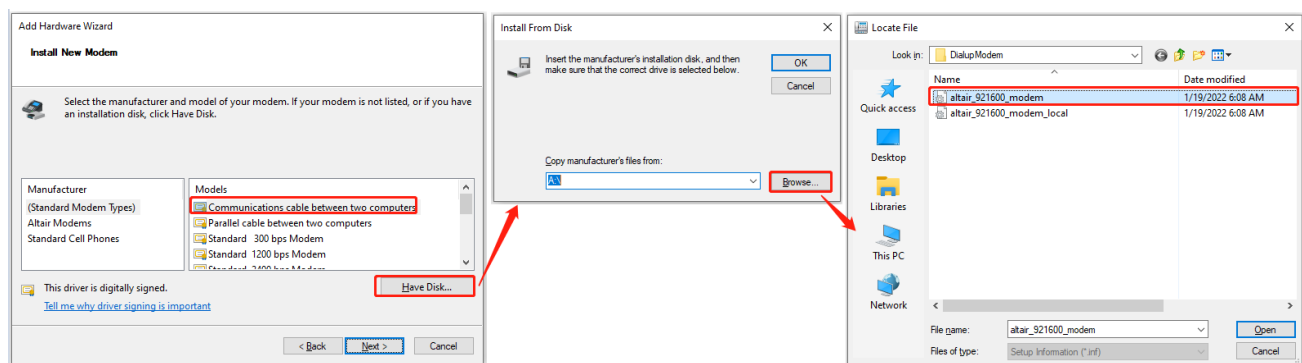


Figure 18: Install 921600 Modem Driver

NOTE

1. The installed RK version must be consistent with the module RK version. Execute **AT+QCFGEXT="VER"** to check the module RK version. For details of the command, see [document \[3\]](#).
2. Modem driver (*altair_921600_modem.inf*) is in RK folder (such as *C:\Program Files (x86)\Altair Semiconductor\RiceKrispies\RK_03_00_00_00_20341_001\Drivers\DialogModem*). Make sure that you have installed Altair RK.

Step 3: Refer to the following three figures to install the new modem according to the instructions on the screen: select "**ALT1250_IoT Modem**", click "**Next**", select "**COM105**", click "**Next**" -> "**Finish**".

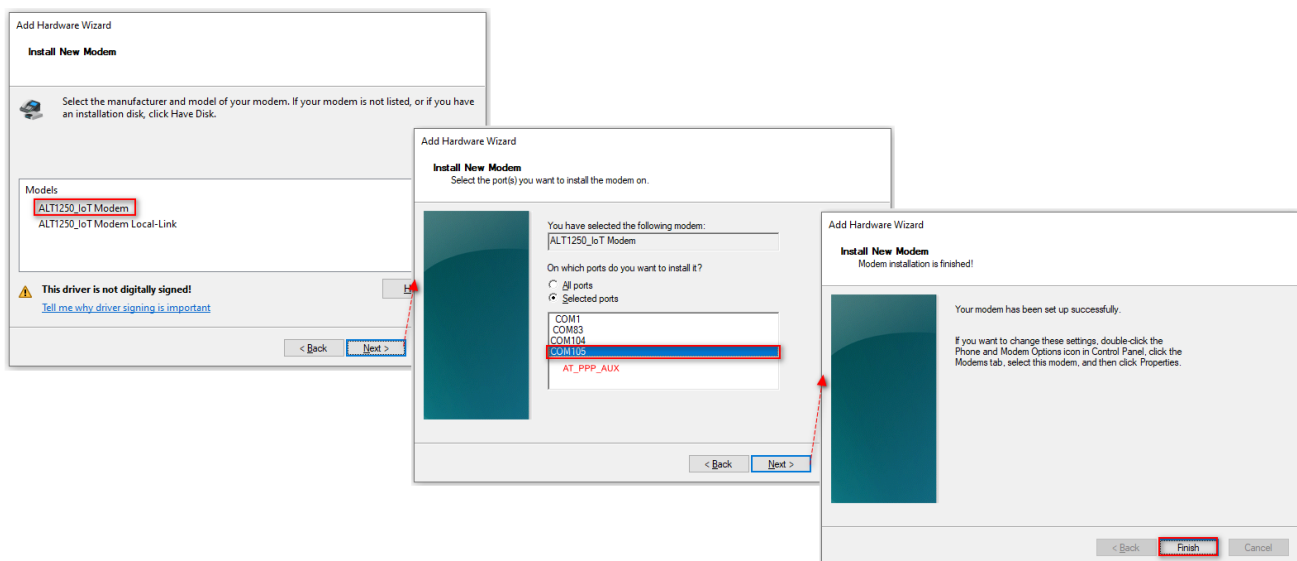


Figure 19: Select a Port to Install the Modem - High-Speed Dial-up

NOTE

If COM105 port is unavailable in Figure 19, make sure COM105 port (AT port) is not connected to a serial port tool.

4.2.2. Configure the Modem Driver

Step 1: Select the "**Communications cable between two computers**" which has been installed; click "**Properties**".

Step 2: Click "**Modem**" to select "**921600**" for "**Maximum Port Speed**"; click "**OK**".

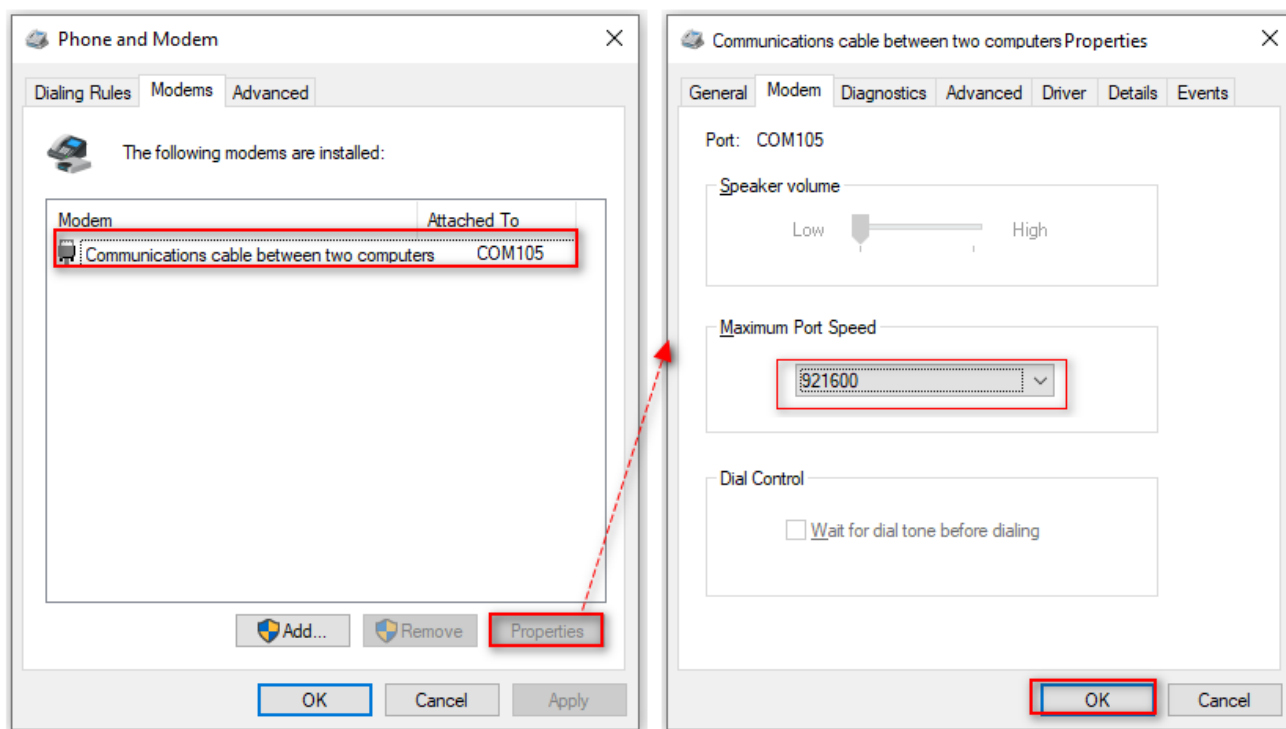


Figure 20: Set Modems Properties - High-Speed Dial-up

NOTE

If there is no "921600" for "Maximum Port Speed", see **Chapter 6.1** for a solution.

4.3. Dial-up Network Configuration

4.3.1. Set up a New Connection

See **Chapter 3.3.1**.

4.3.2. Configure the Connection

See **Chapter 3.3.2**.

4.3.3. Configure the Dial-up Tool

Repeat **Step 1** to **Step 2** in **Chapter 3.3.3** to go to "General" and click "Configure...", select "921600" as "Maximum speed (bps)" and then click "OK".

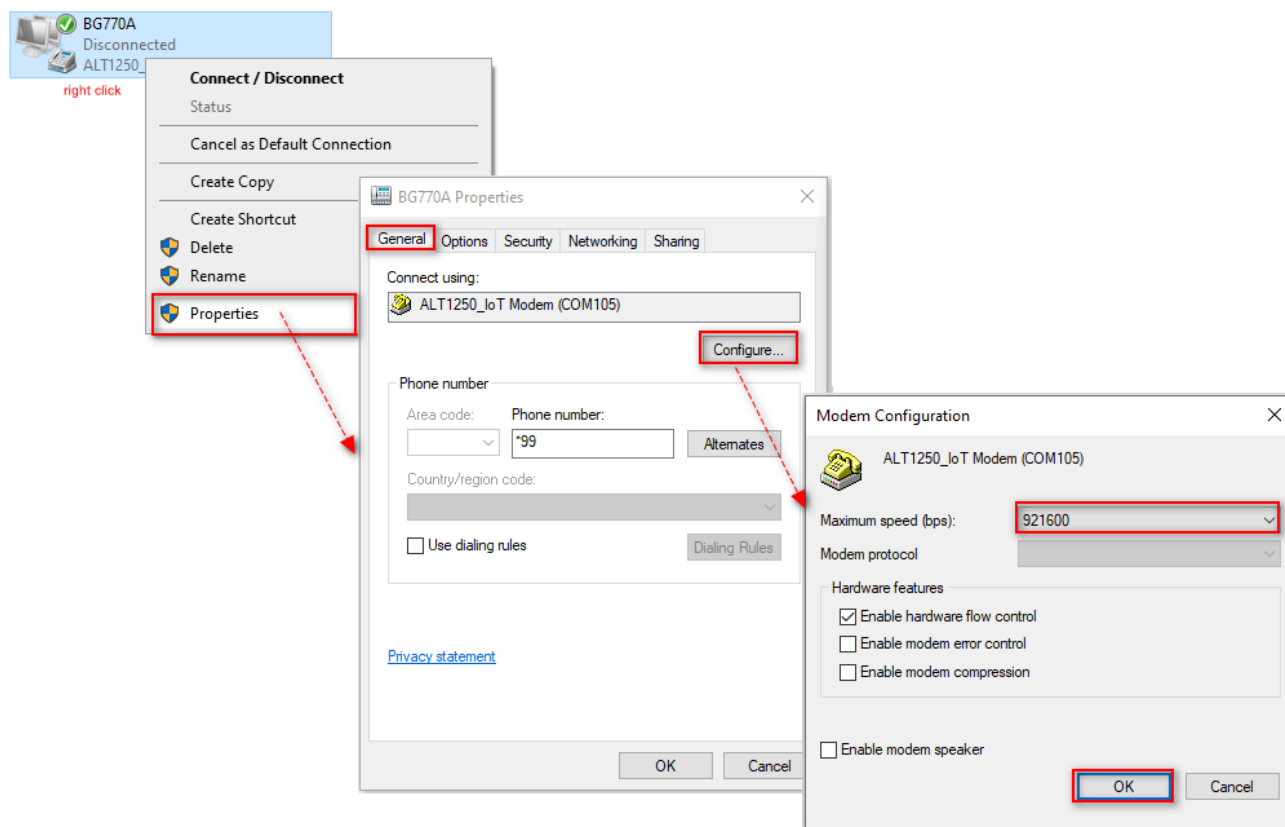


Figure 21: General Settings for Modem Configuration - High-Speed Dial-up

4.4. Establish the Dial-up Connection

See *Chapter 3.4*.

4.5. Restore Default Port Mapping Functions

To restore the default port mapping functions, execute **AT+QCFGEXT="debug",0** at first and then execute **AT+QCFGEXT="PPPMAPPING/SUBPROFILE",0**.

5 Route Configuration

After completing the steps above, PPP can work properly. Many applications on the host send data via PPP, which affects some specific applications. To ensure a valid test, you need to configure the route before testing.

5.1. Query IP Address

Refer to the result querying the IP address with **AT+CGPADDR** on AT tool in **Chapter 3.2.1**, or with **ipconfig** in Windows CMD on the host. For details of **AT+CGPADDR**, see **document** [错误!未找到引用源。](#)

5.2. Delete All Default Routes

Upon a successful PPP connection, input the command below on the Windows CMD immediately to delete all the routes.

```
Route delete 0.0.0.0
```

5.3. Add Target Routes

After deleting the routes, add the target route into the route list to allow the target service. The format is **route add <target address> mask <mask address> <local address>**. For example:

```
route add 220.180.239.212 mask 255.255.255.255 100.82.48.224
```

NOTE

1. 220.180.239.212 in the example is the target address.
2. 100.82.48.224 in the example is the UE address.

6 FAQ

6.1. Disable Driver Signature Enforcement

If there is no 921600 in "Maximum Port Speed" when configuring the modem driver in **Chapter 4.2.2**, as illustrated below, you can fix this problem by referring to the following sections.

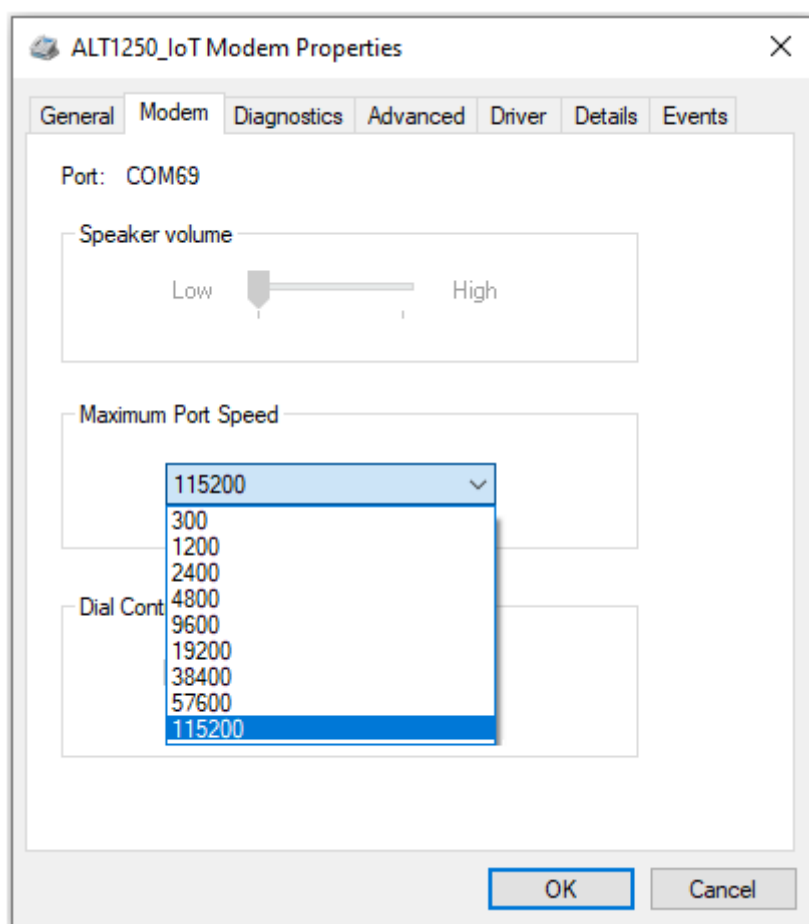


Figure 22: Set Modem Properties

6.1.1. Disable the Driver Signature Enforcement

To disable the driver signature enforcement, open **"Settings"**, click **"Update & Security"** -> **"Recovery"** -> **"Restart Now"** -> **"Troubleshoot"** from **"Choose an Option"** window -> **"Startup Setting"** from **"Advanced Options"** window, and tap **"7"** or **"F7"** on your keyboard, as illustrated below.

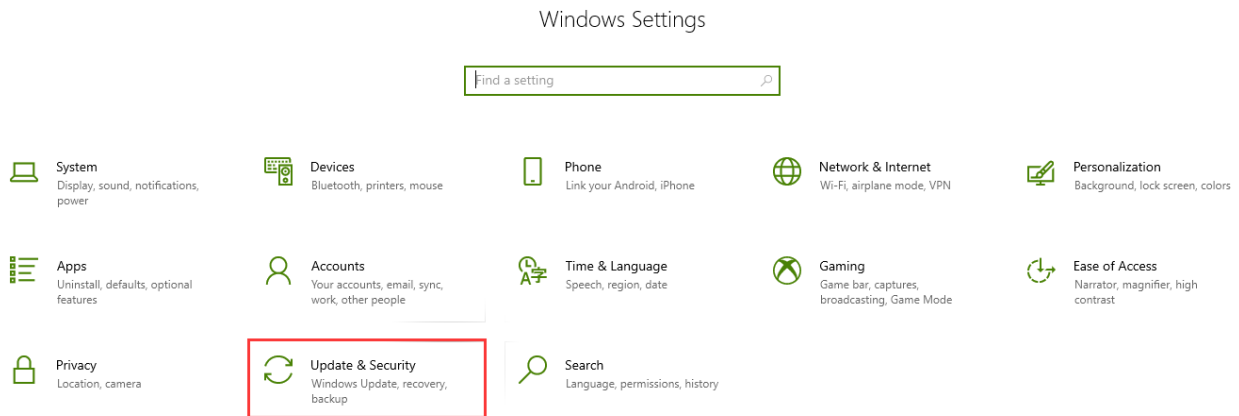


Figure 23: Open "Update & Security"

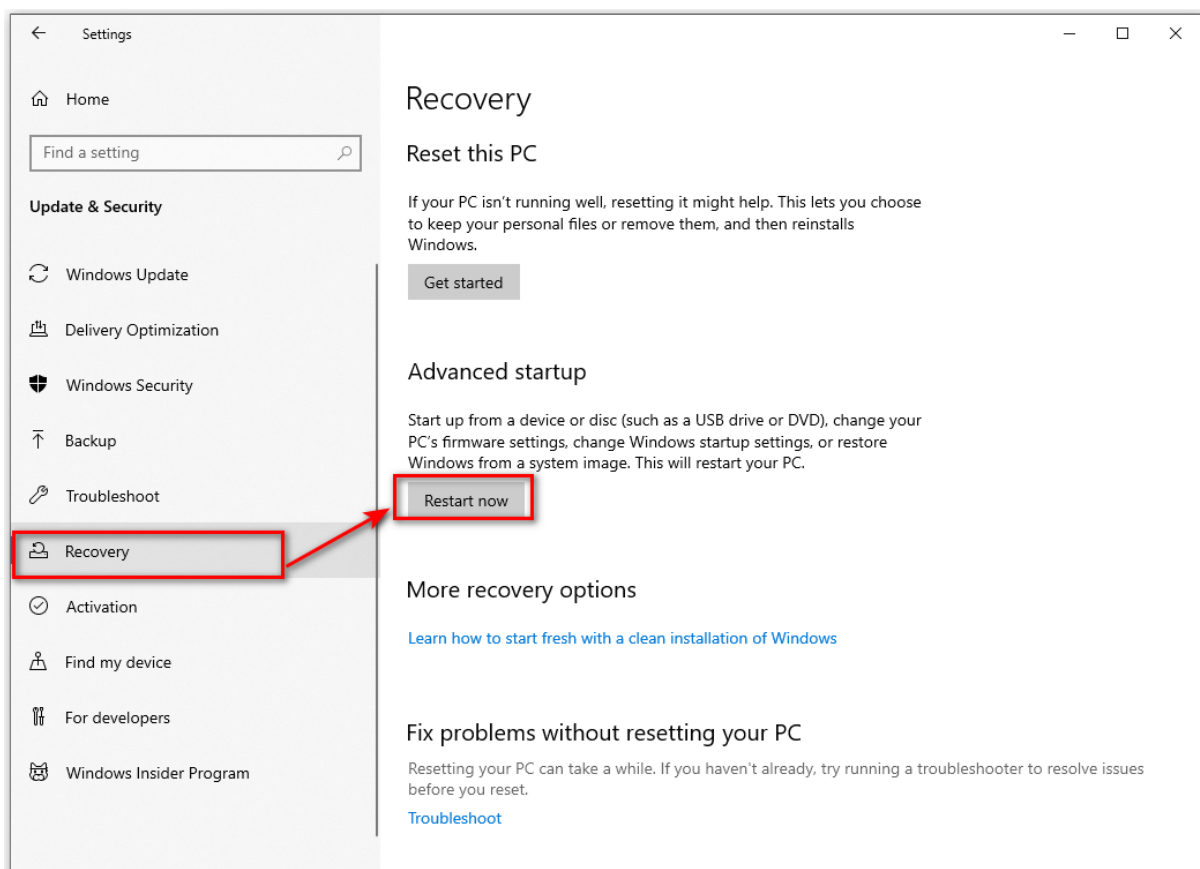


Figure 24: Restart Now

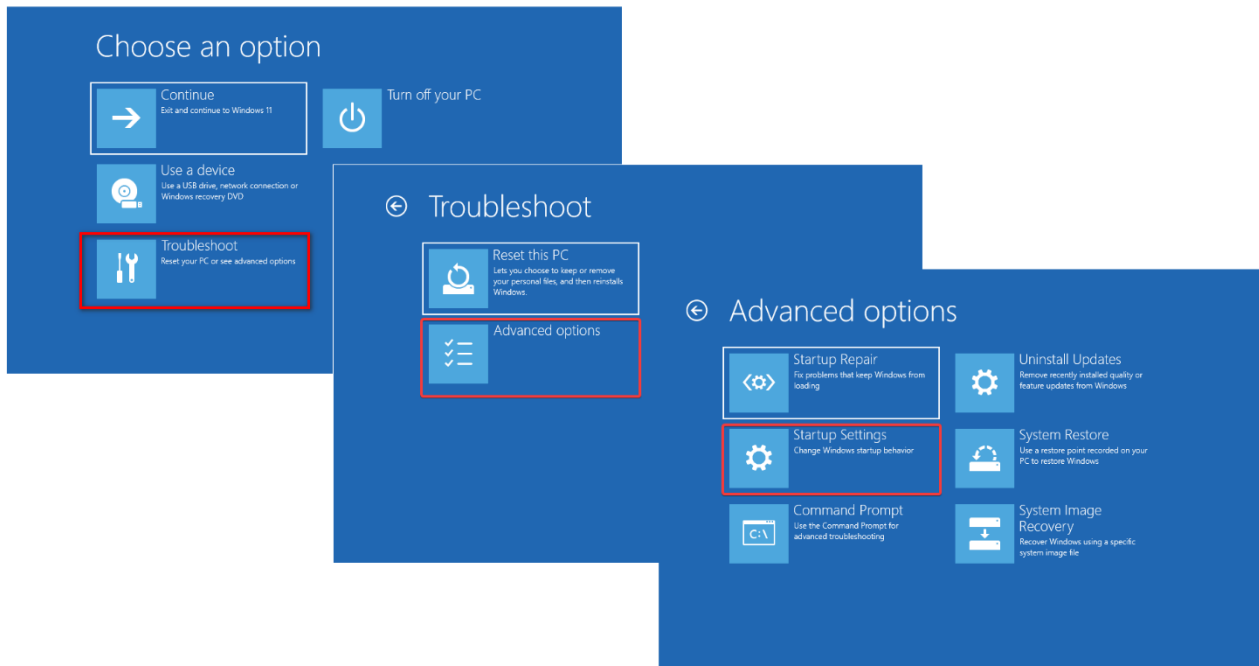


Figure 25: Change Windows Startup Behaviour

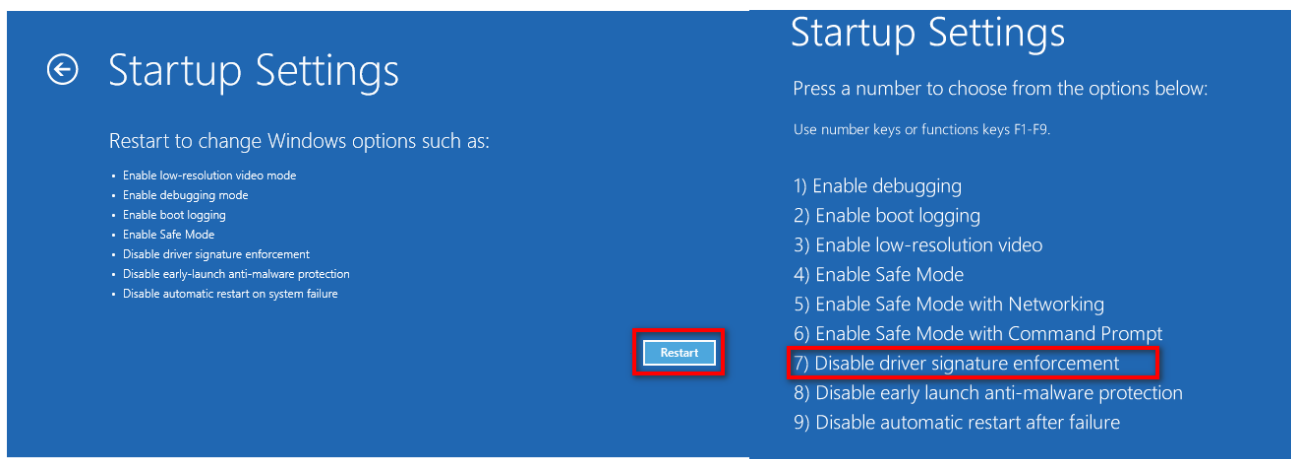


Figure 26: Disable the Driver Signature Enforcement

6.1.2. Update Driver

Repeat **Step 1** in **Chapter 4.2.2** to go to "**Communications cable between two computers**". Then click "**Properties**" -> "**Driver**" to update driver, as illustrated below.

NOTE

Modem driver (*altair_921600_modem.inf*) is in RK folder (such as *C:\Program Files (x86)\Altair Semiconductor\RiceKrispies\RK_03_00_00_00_20341_001\Drivers\DialogModem*). Make sure that you have installed Altair RK.

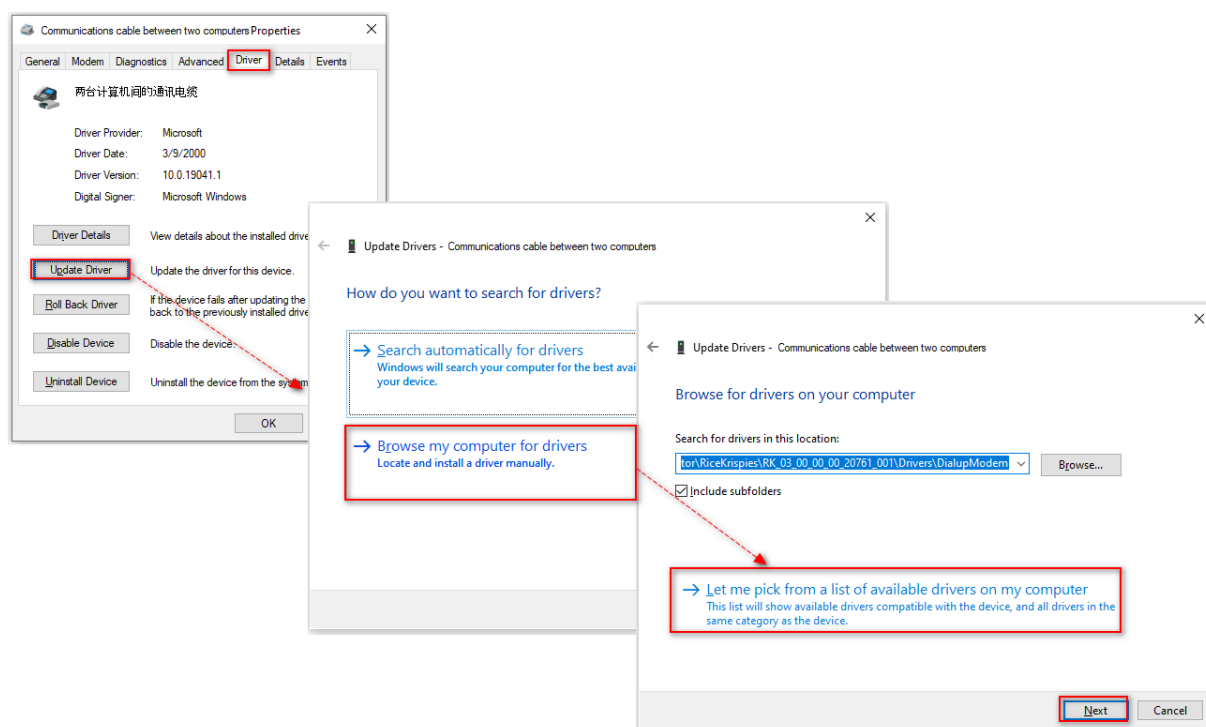


Figure 27: Find Driver Folder Path

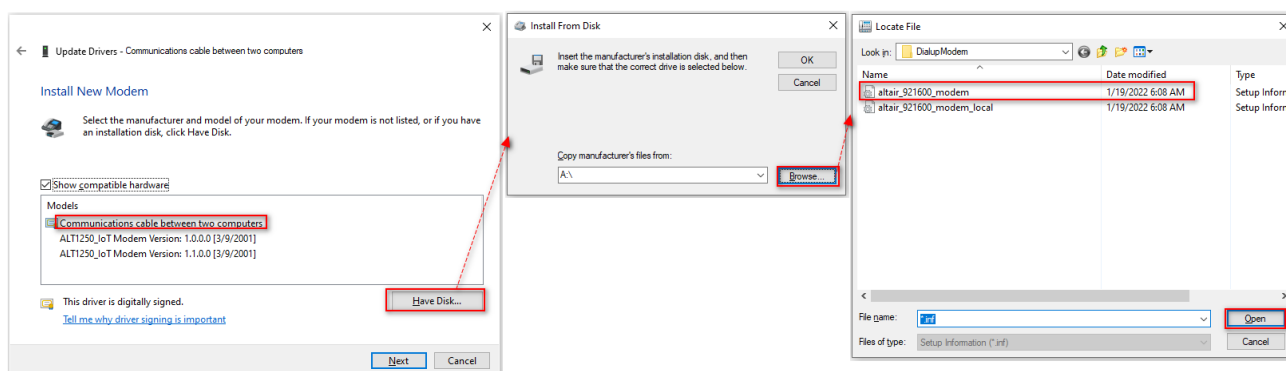


Figure 28: Update Drivers

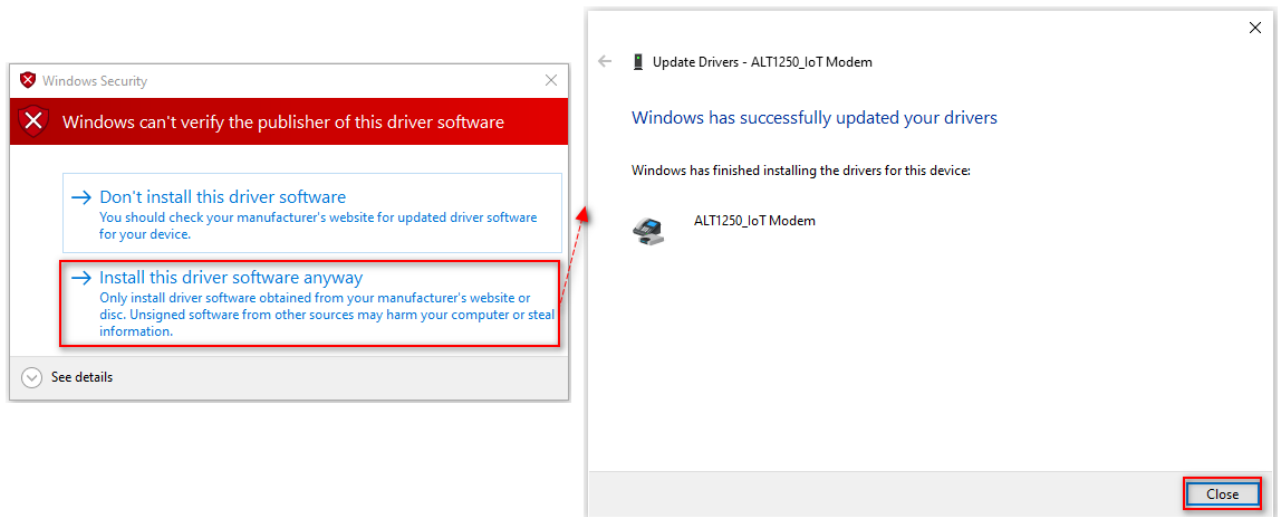


Figure 29: Update Drivers Success

Repeat **Chapter 4.2.2** to configure the Modem driver.

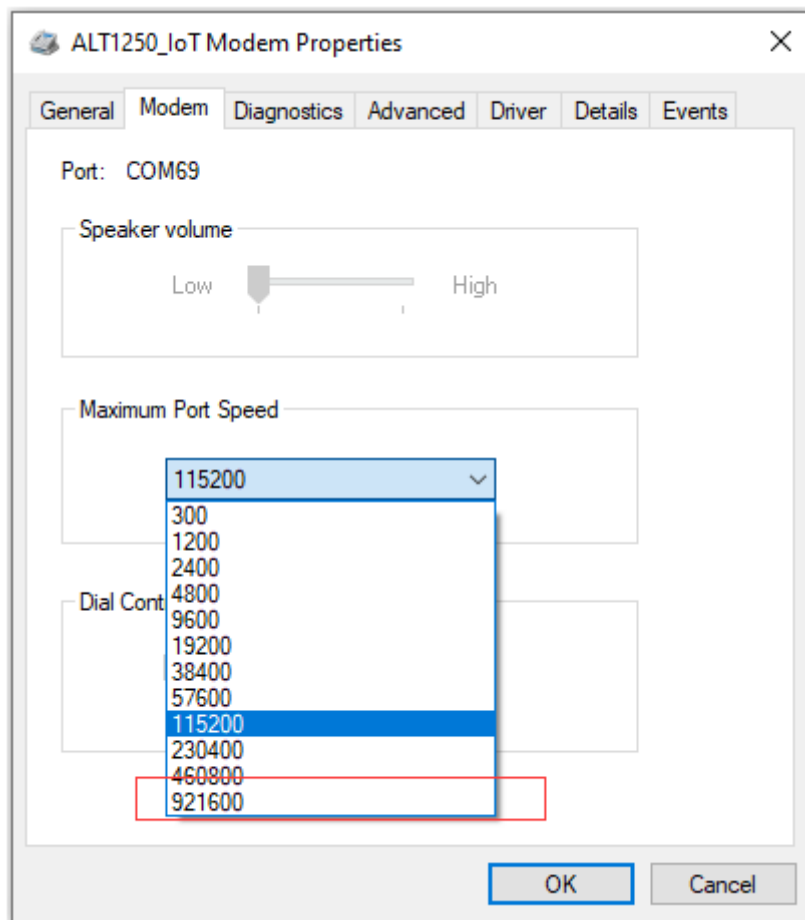


Figure 30: Set Modem Properties

7 Appendix References

Table 5: Related Documents

Document Name
[1] Quectel_BG770A-GL_TE-A_User_Guide
[2] Quectel_BG950A-GL&BG951A-GL_TE-A_User_Guide
[3] Quectel_BG77xA-GL&BG95xA-GL_AT_Commands_Manual
[4] Quectel_BG77xA-GL&BG95xA-GL_QCFGEXT_AT_Commands_Manual
[5] Quectel_BG770A-GL&BG95xA-GL_FTP(S)_Application_Note

Table 6: Terms and Abbreviations

Abbreviation	Description
Cat-M	Category-Minus
COM	Cluster Communication Port
CTS	Clear To Send
EPS	Evolved Packet System
GND	Ground
GNSS	Global Navigation Satellite System
FTP	File Transfer Protocol
LPT	Line Print Terminal
LPWA	Low Power Wide Area
NB-IoT	Narrowband Internet of Things
IP	Internet Protocol

PC	Private Computer
PCM	Pulse Code Modulation
PDP	Packet Data Protocol
PPP	Point-to-Point Protocol
PWR	Power
RF	Radio Frequency
RK	Rice Krispies
RTS	Ready To Send
SD	Secure Digital
UART	Universal Asynchronous Receiver/Transmitter
UE	User Equipment
USB	Universal Serial Bus
USIM	Universal Subscriber Identity Module
VBAT	Voltage of Battery
